

Revision Notes for Class 8 Mathematics

Chapter 1 – Rational Numbers

Rational Numbers:

1. Rational Numbers are numbers in the form of $\frac{p}{q}$ such that $q > 0$. It is denoted by Q.
2. If the numerator and denominator are coprime and $q > 0$ then the Rational Number is of the standard form.

3. Types of Rational Numbers:

- i. **Positive Rational Numbers:** The sign of both the numerator and denominator are the same, i.e., either both are positive, or both are negative. Ex: $\frac{2}{3}, \frac{-7}{-8}, \dots$
- ii. **Negative Rational Numbers:** The sign of both the numerator and denominator are the same, i.e., if the numerator is negative, the denominator will be positive. Similarly, if the numerator is positive, the denominator is negative. Ex: $\frac{2}{-3}, \frac{-7}{8}, \dots$
- iii. **Zero Rational Numbers:** The numerator is always zero. Ex: $\frac{0}{3}, \frac{0}{8}, \dots$

4. Properties of Rational Numbers:

4.1 Closure Property

i. Whole number:

Operation	Numbers	Remarks
Addition	$0+5=5$, a whole number $4+7=11$, a whole number	Whole numbers are closed under addition.
Subtraction	$5-7=-2$, which is not a whole number.	Whole numbers are not closed under subtraction.
Multiplication	$0 \times 3 = 0$, a whole number $3 \times 7 = 21 \dots$	Whole numbers are closed under multiplication.
Division	$5 \div 8 = \frac{5}{8}$, which is not a whole number.	Whole numbers are not closed under division.

ii. Integers

Operation	Numbers	Remarks
Addition	$-6 + 5 = -1$, an integer. In general, $a + b$ is an integer for any two integers a and b	Integers are closed under addition.

Subtraction	$7 - 5 = 2$, an integer. $-6 - 8 = -14$, an integer. $-6 - (-8) = 2$, an integer In general, for any two integers a and b , $a - b$ and $b - a$ is again an integer.	Integers are closed under subtraction.
Multiplication	$5 \times 8 = 40$, an integer. $-5 \times (-8) = 40$, an integer. In general, for any two integers a and b , $a \times b$ is also an integer.	Integers are closed under multiplication.
Division	$5 \div 8 = \frac{5}{8}$, which is not an integer.	Integers are not closed under division.

iii. Rational numbers

To prove the closure property for rational numbers under addition, subtraction, multiplication, and division, let's consider two rational numbers. A rational number can be expressed in the form $\frac{a}{b}$, where a and b are integers, and $b \neq 0$.

1. Addition

To prove that rational numbers are closed under addition:

- Let $\frac{a}{b}$ and $\frac{c}{d}$ be two rational numbers.

- Adding them:

$$\frac{a}{b} + \frac{c}{d} = \frac{ad + bc}{bd}$$

- Since the numerator $ad + bc$ and the denominator bd are both integers (as integers are closed under addition and multiplication), the result is a rational number.

- Conclusion: Rational numbers are closed under addition.

2. Subtraction

To prove that rational numbers are closed under subtraction:

- Let $\frac{a}{b}$ and $\frac{c}{d}$ be two rational numbers.

- Subtracting them:

$$\frac{a}{b} - \frac{c}{d} = \frac{ad - bc}{bd}$$

- Since the numerator $ad - bc$ and the denominator bd are both integers, the result is a rational number.

- Conclusion: Rational numbers are closed under subtraction.

3. Multiplication

To prove that rational numbers are closed under multiplication:

- Let $\frac{a}{b}$ and $\frac{c}{d}$ be two rational numbers.

- Multiplying them:

$$\frac{a}{b} \times \frac{c}{d} = \frac{ac}{bd}$$

- Since the numerator ac and the denominator bd are both integers, the result is a rational number.

- Conclusion: Rational numbers are closed under multiplication.

4. Division

To prove that rational numbers are closed under division (except when dividing by zero):

- Let $\frac{a}{b}$ and $\frac{c}{d}$ be two rational numbers where $c \neq 0$ and $d \neq 0$.

- Dividing them:

$$\frac{a}{b} \div \frac{c}{d} = \frac{a}{b} \times \frac{d}{c} = \frac{ad}{bc}$$

- Since the numerator ad and the denominator bc are both integers and $bc \neq 0$, the result is a rational number.

- Conclusion: Rational numbers are closed under division, except when dividing by zero.

Therefore, rational numbers are closed under addition, subtraction, multiplication, and division (with the exception of division by zero).

4.2 Commutative Property:

i. Whole numbers

Operation	Numbers	Remarks
Addition	$0 + 7 = 7 + 0 = 7$ $2 + 3 = \dots + \dots = \dots$ For any two whole numbers a and b, $a + b = b + a$	Addition is commutative
Subtraction	-	Subtraction is not commutative.
Multiplication	-	Multiplication is commutative.
Division	-	Division is not commutative.

ii. Integers

Operation	Numbers	Remarks
Addition		Addition is commutative.
Subtraction	$5 - (-3) \neq -3 - 5$	Subtraction is not commutative.

Multiplication		Multiplication is commutative.
Division		Division is not commutative.

iii. Rational numbers

Let's now check the commutative property for rational numbers under addition, subtraction, multiplication, and division.

1. Addition (Commutative Property):

The commutative property of addition states that for any two numbers a and b , the order of addition does not affect the result, i.e., $a + b = b + a$.

- Let $\frac{a}{b}$ and $\frac{c}{d}$ be two rational numbers.

- We know:

$$\frac{a}{b} + \frac{c}{d} = \frac{ad + bc}{bd} \quad \text{and} \quad \frac{c}{d} + \frac{a}{b} = \frac{cb + ad}{db}$$

- Since $ad + bc = bc + ad$ and $bd = db$, we conclude that:

$$\frac{a}{b} + \frac{c}{d} = \frac{c}{d} + \frac{a}{b}$$

- Conclusion: Rational numbers are commutative under addition.

2. Subtraction (Commutative Property):

The commutative property of subtraction states that for any two numbers a and b , $a - b$ does not necessarily equal $b - a$.

- Let $\frac{a}{b}$ and $\frac{c}{d}$ be two rational numbers.

- We know:

$$\frac{a}{b} - \frac{c}{d} = \frac{ad - bc}{bd} \quad \text{and} \quad \frac{c}{d} - \frac{a}{b} = \frac{cb - ad}{db}$$

- Since $ad - bc \neq cb - ad$, we conclude:

$$\frac{a}{b} - \frac{c}{d} \neq \frac{c}{d} - \frac{a}{b}$$

- Conclusion: Rational numbers are not commutative under subtraction.

3. Multiplication (Commutative Property):

The commutative property of multiplication states that for any two numbers a and b ,
 $a \times b = b \times a$.

- Let $\frac{a}{b}$ and $\frac{c}{d}$ be two rational numbers.

- We know:

$$\frac{a}{b} \times \frac{c}{d} = \frac{ac}{bd} \quad \text{and} \quad \frac{c}{d} \times \frac{a}{b} = \frac{ca}{db}$$

- Since $ac = ca$ and $bd = db$, we conclude:

$$\frac{a}{b} \times \frac{c}{d} = \frac{c}{d} \times \frac{a}{b}$$

- Conclusion: Rational numbers are commutative under multiplication.

4. Division (Commutative Property):

The commutative property of division states that for any two numbers a and b , $a \div b$ does not necessarily equal $b \div a$.

- Let $\frac{a}{b}$ and $\frac{c}{d}$ be two rational numbers where $c \neq 0$ and $d \neq 0$.

- We know:

$$\frac{a}{b} \div \frac{c}{d} = \frac{ad}{bc} \text{ and } \frac{c}{d} \div \frac{a}{b} = \frac{cb}{ad}$$

- Since $ad \neq cb$ and $bc \neq ad$, we conclude:

$$\frac{a}{b} \div \frac{c}{d} \neq \frac{c}{d} \div \frac{a}{b}$$

- Conclusion: Rational numbers are not commutative under division.

Summary of Commutative Property for Rational Numbers:

- Addition: Commutative.
- Subtraction: Not commutative.
- Multiplication: Commutative.
- Division: Not commutative.

4.3. Associative Property:

i. Whole numbers

Operations	Numbers	Remarks
Addition		Addition is associative
Subtraction		Subtraction is not associative
Multiplication	$7 \times (2 \times 5) = (7 \times 2) \times 5$ $4 \times (6 \times 0) = (4 \times 6) \times 0$ For any three whole numbers a, b and c, $a \times (b \times c) = (a \times b) \times c$	Multiplication is associative
Division		Division is not associative

ii. Integers

Operation	Numbers	Remarks
Addition	For any three integers a, b and c, $a + (b + c) = (a + b) + c$	Addition is associative
Subtraction	$5 - (7 - 3) \neq (5 - 7) - 3$	Subtraction is not associative

Multiplication	For any three integers a, b and c, $a \times (b \times c) = (a \times b) \times c$	Multiplication is associative
Division	$[(-10) \div 2] \div (-5) \neq (-10) \div [2 \div (-5)]$	Division is not associative

iii. Rational Numbers

Let's briefly check the associative property for rational numbers under addition, subtraction, multiplication, and division.

1. Addition (Associative Property):

The associative property of addition states that the grouping of numbers does not affect the sum, i.e., $(a + b) + c = a + (b + c)$.

- Let $\frac{a}{b}$, $\frac{c}{d}$, and $\frac{e}{f}$ be three rational numbers.

- We know:

$$\left(\frac{a}{b} + \frac{c}{d}\right) + \frac{e}{f} = \frac{ad + bc}{bd} + \frac{e}{f} = \frac{(ad + bc)f + bde}{bdf} \quad \text{and}$$

$$\frac{a}{b} + \left(\frac{c}{d} + \frac{e}{f}\right) = \frac{a}{b} + \frac{cf + de}{df} = \frac{(cf + de)b + adf}{bdf}$$

- Both are equal, so:

$$(a + b) + c = a + (b + c)$$

- Conclusion: Rational numbers are associative under addition.

2. Subtraction (Associative Property):

The associative property does not hold for subtraction. This means:

$$(a - b) - c \neq a - (b - c)$$

- Conclusion: Rational numbers are not associative under subtraction.

3. Multiplication (Associative Property):

The associative property of multiplication states that the grouping of numbers does not affect the product, i.e., $(a \times b) \times c = a \times (b \times c)$.

- Let $\frac{a}{b}$, $\frac{c}{d}$, and $\frac{e}{f}$ be three rational numbers.

- We know:

$$\left(\frac{a}{b} \times \frac{c}{d}\right) \times \frac{e}{f} = \frac{ac}{bd} \times \frac{e}{f} = \frac{ace}{bdf} \quad \text{and}$$

$$\frac{a}{b} \times \left(\frac{c}{d} \times \frac{e}{f}\right) = \frac{a}{b} \times \frac{ce}{df} = \frac{ace}{bdf}$$

- Both are equal, so: $(a \times b) \times c = a \times (b \times c)$

- Conclusion: Rational numbers are associative under multiplication.

4. Division (Associative Property):

The associative property does not hold for division. This means:

$$(a \div b) \div c \neq a \div (b \div c)$$

- Conclusion: Rational numbers are not associative under division.

Summary of Associative Property for Rational Numbers:

- Addition: Associative.
- Subtraction: Not associative.
- Multiplication: Associative.
- Division: Not associative.

iv. Distributive Property:

The distributive property links addition and multiplication, stating that multiplication distributed over addition (or subtraction) yields the same result. Mathematically, for any numbers a , b , and c :

$$a \times (b + c) = (a \times b) + (a \times c)$$

This property holds true for integers, whole numbers, and rational numbers.

1. Distributive Property for Integers

Let $a = 3$, $b = -2$, and $c = 4$.

$$3 \times (-2 + 4) = 3 \times 2 = 6$$

On the other hand:

$$(3 \times -2) + (3 \times 4) = -6 + 12 = 6$$

Thus, $3 \times (-2 + 4) = (3 \times -2) + (3 \times 4)$.

- Conclusion: The distributive property holds for integers.

2. Distributive Property for Whole Numbers

Let $a = 2$, $b = 5$, and $c = 3$ (all whole numbers).

$$2 \times (5 + 3) = 2 \times 8 = 16$$

On the other hand:

$$(2 \times 5) + (2 \times 3) = 10 + 6 = 16$$

$$\text{Thus, } 2 \times (5 + 3) = (2 \times 5) + (2 \times 3).$$

- Conclusion: The distributive property holds for whole numbers.

3. Distributive Property for Rational Numbers

$$\text{Let } a = \frac{1}{2}, b = \frac{3}{4}, \text{ and } c = \frac{5}{6}.$$

$$\frac{1}{2} \times \left(\frac{3}{4} + \frac{5}{6} \right) = \frac{1}{2} \times \left(\frac{9}{12} + \frac{10}{12} \right) = \frac{1}{2} \times \frac{19}{12} = \frac{19}{24}$$

On the other hand:

$$\left(\frac{1}{2} \times \frac{3}{4} \right) + \left(\frac{1}{2} \times \frac{5}{6} \right) = \frac{3}{8} + \frac{5}{12} = \frac{9}{24} + \frac{10}{24} = \frac{19}{24}$$

$$\text{Thus, } \frac{1}{2} \times \left(\frac{3}{4} + \frac{5}{6} \right) = \left(\frac{1}{2} \times \frac{3}{4} \right) + \left(\frac{1}{2} \times \frac{5}{6} \right).$$

- Conclusion: The distributive property holds for rational numbers.

Final Summary:

- The distributive property applies to integers, whole numbers, and rational numbers.
- It shows that multiplying a number by a sum (or difference) is the same as multiplying the number by each addend (or subtrahend) and then adding (or subtracting) the results.

v. General Properties:

- A rational number can be a fraction or not, but vice versa is true.
- Rational numbers can be denoted on a number line.
- There are 'n' number of rational numbers between any two rational numbers.

5. Role of Zero: Also known as the Additive Identity

Whenever '0' is added to any rational number, the answer is the Rational number itself.

Ex: If 'a' is any rational number, then $a + 0 = 0 + a = a$

6. Role of One: Also known as the Multiplicative Identity.

Whenever '1' is multiplied by any rational number, the answer is the Rational number itself.

Ex: If 'a' is any rational number, then $a \times 1 = 1 \times a = a$

7. Additive Inverse:

The Additive Inverse of any rational number is the same rational number with the opposite sign. The additive inverse of $\frac{a}{b}$ is $-\frac{a}{b}$. Similarly, the additive inverse of $-\frac{a}{b}$ is $\frac{a}{b}$, where $\frac{a}{b}$ is the rational number.

8. Multiplicative Inverse: Also known as the Reciprocal.

The Multiplicative Inverse of any rational number is the inverse of the same rational number.

The multiplicative inverse of $\frac{a}{b}$ is $\frac{b}{a}$. Similarly, the multiplicative inverse of $\frac{b}{a}$ is $\frac{a}{b}$, where

$\frac{a}{b}$ and $\frac{b}{a}$ is any rational number.